On Ramanujan Challenge Problem 3.2: The Apéry GCD Conjecture

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Abstract

We study $G_n = \gcd(d_n a_n, d_n b_n)$ for the Apéry sequences (a_n), (b_n) with $d_n = \mathrm{lcm}(1, \ldots, n)^3$. We prove *unconditionally* that $G_n = e^{o(n)}$ for a set of positive integers of natural density 1, with the quantitative bound $\#\{n \le N : \log G_n > \varepsilon\, n\} = O(N^{2/3})$ for every $\varepsilon > 0$. The key ingredient is a new bound $Z(p) = O(p^{2/3})$ on the zero-count function $Z(p) = \#\{j < p : p \mid b_j\}$, derived from the Apéry recurrence via a gap polynomial argument. Computation for all 78,496 primes $5 \le p \le 10^6$ gives $Z(p) \le 12$ with mean ≈ 1. The pair count $Z(p)/2$ fits Poisson(1/2); we prove this rigorously in the prime-aspect via a Chebotarev argument for the hyperoctahedral Galois group of the gap polynomials. The Poisson tail predicts $\max_{p \le N} Z(p) \sim 2 \log\log N$, so $Z(p)$ is almost certainly unbounded. Under the weaker Cesàro hypothesis $\sum_{p \le x} Z(p) \ll \pi(x)$ (Hypothesis $\bar{Z}$), which the Poisson model supports, we prove $\log G_n = O(\sqrt{n}(\log n)^{1+\varepsilon})$ for density 1.

1 Setup

The Apéry sequences satisfy the three-term recurrence

$$(n+1)^3 u_{n+1} = (34n^3 + 51n^2 + 27n + 5)\, u_n - n^3\, u_{n-1}, \quad n \ge 1, \tag{1}$$

with $(a_0, a_1) = (0, 6)$ and $(b_0, b_1) = (1, 5)$. Here $b_n = \sum_{k=0}^{n} \binom{n}{k}^2 \binom{n+k}{k}^2 \in \mathbb{Z}$ is the n-th Apéry number. Set $d_n = \mathrm{lcm}(1, \ldots, n)^3$; Apéry [3] showed $d_n a_n, d_n b_n \in \mathbb{Z}$. Define $G_n = \gcd(d_n a_n, d_n b_n)$. Write $P(n) = 34n^3 + 51n^2 + 27n + 5$ for the middle coefficient.

2 The Wronskian identity

Lemma 1. *For all $n \ge 1$, $W_n := a_n b_{n-1} - a_{n-1} b_n = 6/n^3$.*

Proof. The Casorati determinant satisfies $W_{n+1}/W_n = n^3/(n+1)^3$ from (1). Since $W_1 = a_1 b_0 - a_0 b_1 = 6$, telescoping gives $W_n = 6/n^3$. □

Corollary 2. *For every prime $p \ge 5$,*

$$v_p(G_n) \le 3\lfloor \log_p n \rfloor. \tag{2}$$

Proof. G_n divides $d_n(a_n b_{n-1} - a_{n-1} b_n) = 6d_n/n^3$. For $p \ge 5$, $v_p(6) = 0$ and $v_p(d_n/n^3) = 3\lfloor \log_p n \rfloor - 3v_p(n) \le 3\lfloor \log_p n \rfloor$. □

3 Small primes: unconditional bound

Proposition 3. $\sum_{p \le \sqrt{n}} v_p(G_n) \log p = O(\sqrt{n})$.

Proof. By (2), $v_p(G_n) \log p \le 3 \log n$ for every prime $p \ge 5$. Primes $p = 2, 3$ contribute at most $O(\log n)$ each. There are $O(\sqrt{n}/\log n)$ primes up to $\sqrt{n}$, so the sum is at most $3 \log n \cdot O(\sqrt{n}/\log n) = O(\sqrt{n})$. □

4 The multiplicative Lucas congruence

Theorem 4 (Gessel [9]). *For every prime p and $n = \sum_{i=0}^{s} n_i p^i$ in base p,*

$$b_n \equiv \prod_{i=0}^{s} b_{n_i} \pmod{p}. \tag{3}$$

This follows from the combinatorial identity $b_n = \sum_k \binom{n}{k}^2 \binom{n+k}{k}^2$ and termwise application of Lucas's congruence to each binomial coefficient. See also Mimura [13] and McIntosh [12].

5 The zero-count function

Definition 5. For a prime $p \ge 5$, define $Z(p) = \#\{j \in \{0, \ldots, p-1\} : b_j \equiv 0 \pmod{p}\}$.

5.1 Unconditional bound

Lemma 6 (No consecutive zeros). *For every prime $p \ge 5$ and every $j \in \{0, \ldots, p-2\}$, b_j and b_{j+1} cannot both vanish modulo p.*

Proof. Suppose $b_m \equiv b_{m+1} \equiv 0 \pmod{p}$ with $0 \le m \le p-2$. The recurrence (1) at $n = m$ gives $(m+1)^3 b_{m+1} = P(m)\, b_m - m^3\, b_{m-1}$, so $m^3 b_{m-1} \equiv 0 \pmod{p}$. Since $1 \le m \le p-2$ implies $p \nmid m$, we get $b_{m-1} \equiv 0$. Iterating gives $b_0 \equiv 0 \pmod{p}$, contradicting $b_0 = 1$. □

Lemma 7 (Gap polynomial). *Fix $h \ge 2$. Suppose $b_m \equiv 0 \pmod{p}$ and $m + h \le p - 1$. Then*

$$b_{m+h} \equiv C_h(m)\, b_{m+1} \pmod{p}, \tag{4}$$

where $C_h(m)$ is a rational function of m whose numerator (after clearing the denominator $\prod_{j=1}^{h-1}(m+j+1)^3$) is a polynomial of degree exactly $3(h-1)$ with positive leading coefficient.

In particular, if $b_m \equiv b_{m+h} \equiv 0 \pmod{p}$, then by Lemma 6 $b_{m+1} \not\equiv 0$, so $C_h(m) \equiv 0 \pmod{p}$, and m is constrained to at most $3(h-1)$ residue classes modulo p, provided C_h is not identically zero over $\mathbb{F}_p$.

Proof. We iterate the recurrence from the initial condition $(b_m, b_{m+1}) = (0, b_{m+1})$. At step $n = m+1$: $b_{m+2} = P(m+1)\, b_{m+1}/(m+2)^3$, since $b_m = 0$ kills the second term. At step $n = m+j$ for $j \ge 2$: b_{m+j+1} is a $\mathbb{Q}(m)$-linear combination of b_{m+j} and b_{m+j-1}, both already rational multiples of b_{m+1}. The leading coefficient of $P(n)$ is 34 and the subtracted term has leading coefficient 1, so at each step the degree in m increases by 3 and the leading coefficient remains positive. After $h-1$ steps the numerator has degree $3(h-1)$.

For $h = 2$: $C_2(m) = P(m+1)/(m+2)^3$, with numerator $P(m+1) = (2m+3)(17m^2+51m+39)$, a cubic with leading coefficient 34. □

Lemma 8 (Nonvanishing over $\mathbb{F}_p$)**.** *For every prime $p \geq 7$ and every $2 \leq h \leq p$, the numerator of C_h is not the zero polynomial modulo p.*

Proof. Write N_h for the numerator of C_h (a polynomial of degree $3(h-1)$), satisfying $N_{h+1}(m) = P(m+h)\, N_h(m) - (m+h)^6\, N_{h-1}(m)$, with $N_1 = 1$ and $N_2 = P(m+1)$.

First evaluation. Equation (4) at $m = -1$ (where $b_{-1} = 0$, $b_0 = 1$) gives $C_h(-1) = b_{h-1}$, hence $N_h(-1) = b_{h-1} \cdot \big((h-1)!\big)^3$.

Second evaluation. Setting $m = -2$ in the recurrence gives $N_{h+1}(-2) = P(h-2)\, N_h(-2) - (h-2)^6\, N_{h-1}(-2)$, with $N_1(-2) = 1$ and $N_2(-2) = P(-1) = -5$. By induction using the Apéry recurrence at $n = h-2$:

$$N_h(-2) \;=\; -5\, b_{h-2} \big((h-2)!\big)^3 \qquad (h \geq 2). \tag{5}$$

Conclusion. If $N_h \equiv 0 \pmod p$ as a polynomial ($2 \leq h \leq p$, $p \geq 7$), then $N_h(-1) \equiv N_h(-2) \equiv 0$, forcing $b_{h-1} \equiv b_{h-2} \equiv 0 \pmod p$ (since $p \nmid 5$ and $p \nmid (h-1)!, (h-2)!$). These are consecutive zeros, contradicting Lemma 6. □

Proposition 9. *For every prime $p \geq 7$,*

$$Z(p) \leq \left\lfloor \frac{p}{H} \right\rfloor + \tfrac{3}{2}(H-1)(H-2) + 1, \tag{6}$$

where H is any integer with $2 \leq H \leq p$. Optimizing $H = \lfloor (p/3)^{1/3} \rfloor + 1$ gives $Z(p) \leq \frac{3^{4/3}}{2}\, p^{2/3} + O(p^{1/3}) = O(p^{2/3})$ unconditionally for all primes. In particular, $Z(p) = o(p)$.

Proof. Partition $\{0, \ldots, p-1\}$ into $\lceil p/H \rceil$ consecutive blocks of length at most H. In each block, mark the first zero; all other zeros have gap $h \in \{2, \ldots, H-1\}$ to some earlier zero in the same block (gap 1 is impossible by Lemma 6). By Lemma 7, a gap-h pair constrains m to at most $3(h-1)$ residue classes mod p (the denominator factors $(m+j+1)^3$ are nonzero since $0 \leq m+j+1 \leq p-1$). By Lemma 8, $C_h \not\equiv 0$ over $\mathbb{F}_p$, so the numerator roots are at most $3(h-1)$. Summing:

$$Z(p) \leq \left\lceil \frac{p}{H} \right\rceil + \sum_{h=2}^{H-1} 3(h-1) = \frac{p}{H} + \tfrac{3}{2}(H-1)(H-2) + O(1).$$

The minimum over real $H > 0$ occurs at $H = (p/3)^{1/3}$, giving the bound. For $p = 2, 3, 5$: $Z(p) = 0$ (since b_j for $j < 5$ is coprime to $2, 3, 5$). □

Remark 10. The leading coefficient of the numerator of C_h is $U_{h-1}(17)$, the Chebyshev polynomial of the second kind evaluated at 17, satisfying $\ell_1 = 1$, $\ell_{h+1} = 34\ell_h - \ell_{h-1}$. Let $N_h(m)$ denote the cleared numerator of C_h. The identity $P(-u-1) = -P(u)$ (evident from $P(n) = (2n+1)(17n^2 + 17n + 5)$) propagates through the recurrence to give $N_h(-m-h-1) = (-1)^{h-1}\, N_h(m)$, verified computationally for $h \leq 200$. For even h, N_h is an *odd* polynomial in $(m + (h+1)/2)$, forcing the structural factor $(2m+h+1)$. At $h = 2$: factor $(2m+3)$ in $P(m+1) = (2m+3)(17m^2+51m+39)$. The computational data (Section 9) suggests $Z(p) = O(1)$, far stronger than $O(p^{2/3})$.

Remark 11 (Content restriction). A stronger nonvanishing holds: if $p \geq 7$ divides the content of N_h (the gcd of all its coefficients), then $h \geq 2p$. Indeed, $N_h(-r) = (-1)^{r-1} b_{r-1} b_{h-r} \big((r-1)!\big)^3 \big((h-r)!\big)^3$ for $1 \leq r \leq h$ (by induction from the Apéry recurrence); choosing $r = 1, 2$ and using $p \nmid (h-1)!, (h-2)!, 5$ for $h \leq p$ recovers Lemma 8, while the range $p < h < 2p$ is excluded similarly via $r = h - p + 1$. Setting $\beta_n = (n!)^3 b_n$ and $F(z) = \sum_{n \geq 0} b_n z^n$, the endpoint factorization packages into the double generating function $\sum_{r,s \geq 0} \frac{N_{r+s+1}(-r-1)}{(r!)^3 (s!)^3}\, u^r v^s = F(-u)\, F(v)$, showing that the boundary array is a rank-one tensor.

Remark 12 (Adjacent resultant formula). The adjacent resultants satisfy the exact formula

$$\left|\mathrm{Res}(N_h, N_{h+1})\right| = \prod_{j=1}^{h-1} \left(j!^3\, b_j\right)^6. \tag{7}$$

This follows from the recursion $\mathrm{Res}(N_h, N_{h+1}) = N_h(-h)^6\ \mathrm{Res}(N_{h-1}, N_h)$ (since $N_{h+1} \equiv -(m+h)^6 N_{h-1} \pmod{N_h}$), combined with $N_h(-h) = (-1)^{h-1} N_h(-1) = (-1)^{h-1}((h-1)!)^3 b_{h-1}$ (the reflection identity and the first evaluation from Lemma 8). Verified for $h = 2, 3, 4, 5$; the cases $h = 2, 3, 4$ give $\mathrm{Res}(N_2, N_3) = -5^6$, $\mathrm{Res}(N_3, N_4) = -2^{18} \cdot 5^6 \cdot 73^6$, $\mathrm{Res}(N_4, N_5) = 2^{36} \cdot 3^{18} \cdot 5^{12} \cdot 17^{12} \cdot 73^6$.

A key consequence: $p \mid \mathrm{Res}(N_h, N_{h+1})$ if and only if $p \mid b_j$ for some $1 \le j \le h-1$ (when $p > h$, so factorials are coprime to p). In particular, N_2 and N_3 have disjoint root sets modulo p for all $p \ge 7$ (since $5^6 \ne 0$).

Remark 13 (Sharpness of the bound). The exponent $2/3$ in Proposition 9 is optimal for any argument using only the degree bounds $\deg N_h = 3(h-1)$. The exact extremal problem $\max\left\{\sum r_h : r_h \le 3(h-1),\ \sum h\, r_h \le p\right\}$ has value $\frac{3}{2}\, p^{2/3} + O(p^{1/3})$, with leading constant $3/2$ achieved by filling gap sizes $h = 2, 3, \ldots, H$ to capacity and using the remaining space at $h = H+1$, where $H \sim p^{1/3}$. Moreover, degree bounds alone (even using all pairs of zeros, not just consecutive pairs) cannot yield $O(p^{1/2})$: an abstract polynomial model on $\sim p^{2/3}$ points can satisfy $\deg F_h \le 3(h-1)$ for all relevant h. Improving the exponent requires genuinely arithmetic input about the family (N_h).

Remark 14 (Dodgson condensation). The Desnanot–Jacobi identity (Lewis Carroll's condensation) applied to the tridiagonal determinant gives the bilinear relation

$$N_{h+2}(m)\, N_h(m+1) - N_{h+1}(m)\, N_{h+1}(m+1) = -\prod_{j=2}^{h+1} (m+j)^6. \tag{8}$$

This is an identity in $\mathbb{Z}[m]$, verified for $h \le 7$. A consequence: for $m \not\equiv -j \pmod{p}$ $(2 \le j \le h+1)$, the pairs $(m, h+2)$ and $(m+1, h)$ cannot both lie in the zero relation $\mathcal{Z} = \{(m, h) : N_h(m) \equiv 0\}$, since $N_{h+1}(m)\, N_{h+1}(m+1)$ would equal a nonzero product of sixth powers. After a meromorphic gauge $\tau_h(x) = N_h(x)/d_h(x)$ satisfying $d_{h+2}(x)\, d_h(x+1) = \prod_{j=2}^{h+1}(x+j)^6$, this becomes an exact SL_2-tiling: $\tau_{h+1}(x)\, \tau_{h+1}(x+1) - \tau_{h+2}(x)\, \tau_h(x+1) = 1$, equivalently an A_1 T-system (discrete Liouville equation). A zero of τ forces its transverse neighbors to be units (*singularity confinement*), but abstract SL_2-tilings can have zeros of density $1/2$ (e.g., $q_n = 0, 1, 0, -1, \ldots$ with $q_n^2 - q_{n-1} q_{n+1} = 1$), so the bilinear structure alone does not force a sublinear zero count. The arithmetic content—endpoint factorization, restart identity (Lemma 20), and polynomial degree bounds—is essential.

Remark 15 (Projective orbit reformulation). Let $B_n = n!^3\, b_n$ and D_n be the second solution of the renormalized recurrence $w_{n+1} = P(n)\, w_n - n^6\, w_{n-1}$ with $(D_0, D_1) = (0, 1)$. Their Casoratian is $B_n D_{n+1} - D_n B_{n+1} = (n!)^6$, nonzero for $n < p$. Define the *projective Apéry orbit* $\pi\colon \{0, \ldots, p-1\} \to \mathbb{P}^1(\mathbb{F}_p)$ by $\pi_n = [B_n : D_n]$. For $0 \le m < m+h < p$, $N_h(m) \equiv 0 \pmod{p}$ if and only if $\pi_m = \pi_{m+h}$. Thus the zero-count $Z(p)$ equals the number of distinct collisions in the orbit π. The reflection symmetry $P(-1-n) = -P(n)$ implies $b_{p-1-n} \equiv b_n \pmod{p}$, making the orbit a palindrome: $\pi_n = \pi_{p-1-n}$. In particular, $Z(p)$ is even except at the non-ordinary primes $p \mid a_p(f)$ (at most two primes $p \le 10^5$).

Remark 16 (Galois structure). For each fixed h, the Galois group of the splitting field of N_h over $\mathbb{Q}$ is not the full symmetric group: the reflection $N_h(-m-h-1) = (-1)^{h-1} N_h(m)$ forces roots into pairs $\{r, -r-h-1\}$, and $\mathrm{Gal}(N_h) \cong C_2 \wr S_m$ (hyperoctahedral group) with $m = \lfloor 3(h-1)/2 \rfloor$ (confirmed exactly for $h \le 5$, $2 \le h \le 9$ by obstruction tests). By the Chebotarev density theorem, the average number of $\mathbb{F}_p$-roots of N_h is 1 for odd h and 2 for even h. This prediction is verified: for 262 primes in $[1000, 3000]$, the measured averages are $0.98, 2.03, 1.11, 2.07, 1.11, 1.88$ for $h = 3, \ldots, 8$.

Proposition 17 (Hyperoctahedral Poisson law)**.** *Assume* $\mathrm{Gal}(N_h) \cong C_2 \wr S_m$ *with* $m = \lfloor 3(h-1)/2 \rfloor$. *Let* $Y_h(p)$ *be the number of non-central reflection-pairs of roots of* N_h *that split completely over* $\mathbb{F}_p$. *Then for each fixed* $k \le m$, *the factorial moments in the prime-aspect satisfy*

$$\lim_{x\to\infty} \frac{1}{\pi(x)} \sum_{p\le x} (Y_h(p))_k = \Big(\frac{1}{2}\Big)^k,$$

where $(Y)_k = Y(Y-1)\cdots(Y-k+1)$. *Since these match the factorial moments of* Poisson$(1/2)$ *for all* k, *the prime-aspect distribution of* $Y_h(p)$ *converges to* Poisson$(1/2)$ *as* $h \to \infty$.

Proof. In $G = C_2 \wr S_m = \{(\sigma,\varepsilon) : \sigma \in S_m,\ \varepsilon \in \{\pm 1\}^m\}$, a *positive fixed point* of $g = (\sigma, \varepsilon)$ is an index i with $\sigma(i) = i$ and $\varepsilon_i = +1$. The Frobenius at p acts on the m reflection-pairs, and $Y_h(p)$ equals the number of positive fixed points. For any k distinct indices $i_1, \ldots, i_k$, the fraction of $g \in G$ fixing all k positively is

$$\frac{2^{m-k}\,(m-k)!}{2^m\, m!} = \frac{1}{2^k\,(m)_k}.$$

Summing over all $(m)_k$ ordered k-tuples, $\mathbf{E}_{g\in G}[(Y(g))_k] = (1/2)^k$, which matches the factorial moments of Poisson$(1/2)$. The Chebotarev density theorem then gives the prime-aspect convergence (Remark 19 supplies the effective error term under GRH). Since the factorial moments determine the Poisson distribution and they hold for all $k \le m$, $Y_h(p) \to$ Poisson$(1/2)$ as $m \to \infty$. □

Remark 18 (Prime-aspect vs. orbit-aspect)*.* Proposition 17 is a *prime-aspect* statement: for fixed h and varying p, the root-pair count $Y_h(p)$ converges to Poisson$(1/2)$. The zero-count $Z(p)$ aggregates contributions from all gaps $h = 1, \ldots, p-1$ at a *fixed* prime p, so the orbit-aspect independence needed to deduce $Z(p)/2 \sim$ Poisson$(1/2)$ is not implied by Chebotarev alone. The gap between the two aspects parallels the gap between the Sato–Tate distribution (average over primes) and pointwise distribution (a single prime).

Remark 19 (GRH-effective error)*.* For each fixed k, the k-th factorial moment does not require the full splitting field K_h (of degree $2^m m!$, double-exponential in h). Let $E_{h,k}$ be the fixed field of the stabilizer of an ordered signed k-tuple with distinct underlying reflection pairs; then $[E_{h,k} : \mathbb{Q}] = 2^k\,(m)_k = O_k(h^k)$. Under GRH for $\zeta_{E_{h,k}}$,

$$\frac{1}{\pi(x)} \sum_{\substack{p\le x\\ p \text{ good}}} (Y_h(p))_k = 2^{-k} + O_k\!\left(\frac{\log D_{h,k} + 2^k (m)_k \log x}{\sqrt{x}}\right),$$

where $D_{h,k} = |\operatorname{Disc}(E_{h,k})|$. The Apéry recurrence gives the coefficient-height bound $\log \|N_h\|_1 = O(h \log h)$, from which $\log D_{h,k} \ll_k h^{k+2} \log h$. Thus an $O(h^{-C})$ error holds in the polynomial range $x \ge h^{2k+4+2C+\varepsilon}$, for each fixed k.

At $x = 10^6$ with $h = 10$ ($m = 13$), the degree contributions alone for $k = 1, 2, 3$ are 0.25, 5.99, and 131.7 respectively: the observed Poisson accuracy is far better than present effective Chebotarev can certify.

5.2 Dual bounds: the restart identity and column estimate

The bound $Z(p) = O(p^{2/3})$ controls zeros along a *fixed row* (varying m for a given gap h). A dual bound controls zeros along a *fixed column* (varying h for a given starting point x_0).

Lemma 20 (Restart identity)**.** *Suppose $N_r(x_0) = 0$ and $N_{r+1}(x_0) \neq 0$ for some $x_0 \in \mathbb{F}_p$ and $r \geq 1$. Then for every $d \geq 0$,*

$$N_{r+d}(x_0) = N_{r+1}(x_0) \cdot N_d(x_0 + r). \tag{9}$$

In particular, if $N_{r+d}(x_0) = 0$ as well, then $N_d(x_0 + r) = 0$.

Proof. The sequence $Y_d = N_{r+d}(x_0)/N_{r+1}(x_0)$ has $Y_0 = 0$, $Y_1 = 1$, and satisfies $Y_{d+1} = P((x_0+r)+d)\, Y_d - ((x_0+r)+d)^6\, Y_{d-1}$, which is the defining recurrence for $N_d(x_0 + r)$. □

Proposition 21 (Column bound)**.** *For every $x_0 \in \mathbb{F}_p$ and every interval I of H consecutive h-values with $H \leq p$,*

$$\#\{h \in I : N_h(x_0) \equiv 0 \pmod{p}\} \leq \tfrac{3^{4/3}}{2}\, H^{2/3} + O(H^{1/3} + 1). \tag{10}$$

Proof. Split I at the unique transition h where $x_0 + h \equiv 0 \pmod{p}$ (if it exists in I), creating at most two nonsingular pieces plus an $O(1)$ additive cost.

On a nonsingular piece, order the zero levels. For threshold $L \geq 2$: a gap of length $d \leq L$ between consecutive zeros at levels $r < r + d$ gives $N_d(x_0 + r) = 0$ by Lemma 20. For fixed d, there are at most $3(d{-}1)$ values of $x_0 + r$ satisfying this (by Lemma 8 and $\deg N_d = 3(d{-}1)$). Gaps longer than L contribute at most $H/(L{+}1)$ zeros. Summing:

$$\text{zero count} \leq \frac{H}{L+1} + \sum_{d=2}^{L} 3(d{-}1) + O(1) = \frac{H}{L+1} + \tfrac{3}{2}L(L{-}1) + O(1).$$

Optimizing $L \asymp (H/3)^{1/3}$ gives the stated bound. □

Computationally, the column count at mesoscopic scale $H = \lfloor\sqrt{p}\rfloor$ is far below the $O(H^{2/3})$ bound: for all primes $p \leq 100{,}003$ and all x_0, $\#\{h \leq H : N_h(x_0) \equiv 0\} \leq 2$. For $p \geq 10{,}007$, the maximum is ≤ 1.

Remark 22 (Rectangular incidence estimate). The row bound $|R_h| \leq 3(h{-}1)$ and the column bound (10) together give the rectangular incidence estimate

$$\#\{(x, h) : x \in \mathbb{F}_p,\ 2 \leq h \leq H,\ N_h(x) \equiv 0\} \leq \min\{H^2,\ pH^{2/3}\}.$$

The first bound sums row degrees; the second sums column estimates. They cross at $H \sim p^{3/4}$. Note that a uniform bound $R_p(H) \leq CH$ for all primes is *false*: by the Chebotarev density theorem, for each fixed H, infinitely many primes $p > H^2$ have every N_h ($2 \leq h \leq H$) splitting completely over $\mathbb{F}_p$, giving $R_p(H) = \frac{3}{2}H(H-1)$. The correct target is the *mesoscopic* estimate $R_p(\lfloor\sqrt{p}\rfloor) = O(\sqrt{p})$, where the split-prime obstruction does not apply because $H = \sqrt{p}$ grows with p. More generally, if $R_p(H) \ll H^\beta$ then $Z(p) \leq p/H + R_p(H)$ is optimized at $H \sim p^{1/(\beta+1)}$, giving $Z(p) \ll p^{\beta/(\beta+1)}$: the current $\beta = 2$ gives $2/3$ and $\beta = 1$ yields $1/2$. Reflection forces $R_p(H) \geq H/2$ from central roots, so $\beta = 1$ is the intrinsic limit of this first-moment method.

5.3 The continuant Bezout identity

Write $S_d(x) = \prod_{j=2}^{d}(x + j)$.

Proposition 23 (Bezout identity)**.** *For integers $2 \leq d < e$,*

$$N_{e-1}(x{+}1)\, N_d(x) - N_{d-1}(x{+}1)\, N_e(x) = S_d(x)^6\, N_{e-d}(x + d). \tag{11}$$

In particular, after saturating the cut-edge divisor S_d,

$$(N_d, N_e) : S_d^\infty = (N_d,\ N_{e-d}(x + d)) : S_d^\infty.$$

Proof. The continuant addition formula (splitting the interval $[1, e-1]$ at position d) gives $N_e = N_d\, N_{e-d+1}(x+d-1) - (x+d)^6\, N_{d-1}\, N_{e-d}(x+d)$. The Dodgson identity at index $d-2$ gives $N_{d-1}(x+1)\, N_{d-1} - N_{d-2}(x+1)\, N_d = \prod_{j=2}^{d-1}(x+j)^6$. Combining these two identities yields (11). Verified computationally for all $2 \le d < e \le 9$. $\square$

Corollary 24 (Separated-block criterion)**.** *Away from the cut-edge divisor* $S_d(x) = 0$*,*

$$N_d(\alpha) \equiv N_e(\alpha) \equiv 0 \pmod p \quad\Longleftrightarrow\quad N_d(\alpha) \equiv N_{e-d}(\alpha + d) \equiv 0 \pmod p.$$

In particular, $\deg \gcd_{\mathbb{F}_p}(N_d, N_e) \le 3(\min(d, e-d) - 1)$*, which improves the naive* $3(d-1)$ *when* $e - d < d$*.*

Remark 25 (Separated-block resultant)*.* Define $\mathcal{R}_{d,r} = \mathrm{Res}_x(N_d(x), N_r(x+d))$. The Bezout identity gives

$$|\operatorname{Res}(N_d, N_e)| = \prod_{j=1}^{d-1} \big((j!)^3 b_j\big)^6 \cdot |\mathcal{R}_{d,e-d}|.$$

For adjacent indices ($e = d+1$), $N_1 = 1$ gives $\mathcal{R}_{d,1} = 1$, recovering the adjacent resultant formula. For nonadjacent indices, $\mathcal{R}_{d,r}$ is genuinely new. The reflection identity gives $|\mathcal{R}_{d,r}| = |\mathcal{R}_{r,d}|$. The subtraction step does *not iterate*: after one application, $N_d(x)$ and $N_{e-d}(x + d)$ live on disjoint shifted intervals with opposite orientations, so no second nested division is possible. For $d = 2$, writing $P(y) = (2y+1)Q(y)$ with $Q(y) = 17y^2+17y+5$ and θ a root of Q, the norm decomposition $\mathcal{R}_{2,r} = 2^{3(r-1)} G_r(-\frac{1}{2}) \cdot 17^{3(r-1)}\, \mathrm{N}_{\mathbb{Q}(\theta)/\mathbb{Q}}(G_r(\theta))$ (where $G_r = N_r(x+1)$) gives $\mathcal{R}_{2,2} = 2^5 \cdot 5^4 \cdot 11^2 \cdot 17^3 \cdot 71$. At $p = 71$: $N_2(-5/2) \equiv N_4(-5/2) \equiv 0 \pmod{71}$ with $-5/2 \notin \{-1, -2, -3, -4\}$, a nonboundary common root inside the mesoscopic range ($4^3 < 71$). This shows the boundary-only gcd statement is *false*.

Remark 26 (Collision energy and the $H^{1/2}$ barrier)*.* The column exponent $2/3$ is the natural barrier for the consecutive-gap argument: even if each gap of length d has only d (rather than $3(d-1)$) admissible starts, the sum $\sum_{d \le L} d \asymp L^2$ still gives $H/L + L^2$, optimized at $H^{2/3}$. To achieve $H^{1/2}$, one needs the aggregate collision energy $\mathcal{E}_p(H) = \sum_{d+r \le H} \#\{x : N_d(x) \equiv N_r(x+d) \equiv 0\} \ll H^{3/2}$, which is a sparse-exception theorem for the separated-block resultant family—beyond the scope of universal continuant identities. (The A_1 T-system $\tau_{h+1}(x)\tau_{h+1}(x+1) - \tau_{h+2}(x)\tau_h(x+1) = 1$ admits tame solutions with zero density 1/2, e.g. $\tau_{2k} = 0$, $\tau_{2k+1} = (-1)^k$, so singularity confinement alone cannot force sparsity.) In the transfer-matrix formulation (writing $M(n) = \left(\begin{smallmatrix} P(n) & -n^6 \\ 1 & 0 \end{smallmatrix}\right)$, so that $N_h(x) = 0$ iff a fixed matrix coefficient of the interval product $M(x+h-1) \cdots M(x+1)$ vanishes), the conjectured $\mathcal{E}_p(H) \ll H^{3/2}$ is analogous to a Rudnev-type point–plane incidence bound in $\mathrm{PGL}_2(\mathbb{F}_p)$, subject to a nonconcentration hypothesis for the deterministic Apéry cocycle. Computation for selected primes:

p	71	503	2003	5003	7919	20011	50021	10^5+3
$H = \lfloor\sqrt{p}\rfloor$	8	22	44	70	88	141	223	316
$R_p(H)/H$	1.75	1.59	1.36	1.50	1.75	1.63	1.56	1.46
$\mathcal{E}_p(H)$	1	2	2	2	10	0	2	0
$\mathcal{E}_p(H)/H^{3/2}$	.044	.019	.007	.003	.012	0	.001	0

The ratio $R_p(H)/H$ clusters around 3/2 (the Chebotarev prediction from Remark 16), and $\mathcal{E}_p(H)/H^{3/2} \to 0$, consistent with $\mathcal{E}_p(H) \ll H^{3/2}$. The key arithmetic object is the shifted resultant $S_{d,r} = \mathrm{Res}_x(N_d(x), N_r(x+d)) \in \mathbb{Z}$: for $p \nmid S_{d,r}$, the gap polynomials N_d and $N_r(\cdot+d)$ share no root over $\mathbb{F}_p$, so that gap pair contributes zero to $\mathcal{E}_p$. A Hadamard bound gives $\log|S_{d,r}| \ll (d+r)^2 \log(d+r)$, so the total number of prime divisors across all pairs (d, r) with $d + r \le H$ is $O(H^4 \log H)$. Averaging over the $\sim X/\log X$ primes in $(X, 2X]$ gives $\mathcal{E}_p(H) \ll H^{1/2}$ for all but $o(X/\log X)$ primes, provided $H \le X^{2/7-\varepsilon}$—rigorous but far from the global target $H = p - 1$.

A sharper route uses the bipartite incidence graph $\Gamma_p(H)$: left vertices are gap lengths d, right vertices are gap lengths r, with an edge when N_d and $N_r(\cdot + d)$ share a root over $\mathbb{F}_p$. Each edge certifies a triple collision, so $\mathcal{E}_p(H) = e(\Gamma_p(H))$. A uniform fiber bound $\deg_L(d) \le C$ (the number of r-partners for each d) and a uniform codegree bound (at most C common right neighbors for any two left vertices) would place $\Gamma_p(H)$ in the regime of the Kővári–Sós–Turán theorem, giving $\mathcal{E}_p(H) \ll H^{3/2}$—the collision-energy scale needed for the mesoscopic target $R_p(\sqrt{p}) = O(\sqrt{p})$. The Dodgson identity (Remark 14) restricts local configurations in Γ_p, but by itself controls $O(H)$ pairs out of $O(H^2)$; upgrading it to a global codegree bound is the key missing step.

5.4 Computational evidence

Proposition 27. *For all* 78,496 *primes* $5 \le p \le 10^6$*:*

1. $Z(p) \in \{0, 2, 4, 6, 8, 10, 12\}$ *for all but two primes; the two exceptions ($p = 11$ and $p = 3137$) have $Z(p) = 1$ (non-ordinary primes, where $p \mid a_p(f)$).* $\max Z(p) = 12$ *at* $p = 159{,}977$.

2. *The exact histogram is:*

$Z(p)$	*0*	*1*	*2*	*4*	*6*	*8*	*10*	*12*
count	*47632*	*2*	*23730*	*6045*	*951*	*123*	*12*	*1*
fraction	*.607*	*.000*	*.302*	*.077*	*.012*	*.002*	*.000*	*.000*
Poi$(\frac{1}{2})$	*.607*	—	*.303*	*.076*	*.013*	*.002*	*.000*	*.000*

The mean $Z(p) = 1.000$, stable across all ranges ($\chi^2 = 3.43$ on 5 d.f. against Poisson(1/2) *for the pair count).*

3. *The symmetry $b_j \equiv b_{p-1-j} \pmod{p}$ holds for all tested primes: zeros come in palindromic pairs $\{j, p-1-j\}$, so $Z(p)$ is even (except at non-ordinary primes). The pair count $Z(p)/2$ fits* Poisson(1/2) *to within sampling noise (table above). Proposition 17 proves* Poisson(1/2) *rigorously in the prime-aspect; the orbit-aspect fit (Remark 18) remains empirical.*

Remark 28 (Mesoscopic root count). At the scale $H = \lfloor\sqrt{p}\rfloor$, the total root count $R_p(H) = \sum_{h=2}^{H} r_p(h)$ satisfies $R_p(H)/H \approx 3/2$ for all tested primes, consistent with the Chebotarev prediction from Remark 16:

p	$H = \lfloor\sqrt{p}\rfloor$	$R_p(H)$	$R_p(H)/H$
503	22	35	1.59
2003	44	60	1.36
5003	70	105	1.50
10007	100	154	1.54
20011	141	230	1.63
50021	223	347	1.56
100003	316	460	1.46

The ratio $R_p(H)/H$ stabilizes near 3/2 (range 1.32–1.75 for $p > 500$), supporting the mesoscopic conjecture in Remark 22.

5.5 Hypothesis Z

Hypothesis 29 (Hypothesis Z)**.** *There exists an absolute constant C such that $Z(p) \le C$ for all primes p.*

Remark 30. The analogous problem for the $\zeta(2)$ Apéry numbers $A_n = \sum_k \binom{n}{k}^2 \binom{n+k}{k}$ has a fundamentally different structure. Stienstra–Beukers [16] showed that A_p is related to a *CM* modular form, yielding $A_{(p-1)/2} \equiv 0 \pmod{p}$ for all $p \equiv 3 \pmod{4}$ — a *positive density* of primes with guaranteed zeros. In contrast, the $\zeta(3)$ Apéry numbers are governed by the non-CM newform $f(z) = \eta(2z)^4\eta(4z)^4 \in S_4(\Gamma_0(8))$ (LMFDB label 8.4.a.a), for which no such systematic zero-forcing occurs. This is a structural reason why Hypothesis $\bar{Z}$ should hold for the $\zeta(3)$ sequence, even though the pointwise bound $Z(p) \le C$ (Hypothesis Z) likely fails.

Remark 31. Even the single-row slice $Z^*(p) := \mathbf{1}[p \mid b_{(p-1)/2}]$ is connected to the non-ordinary primes of f via the Beukers congruence $b_{(p-1)/2} \equiv a_p(f) \pmod{p}$ [5, 1]. The density-zero conjecture for non-ordinary primes of non-CM weight-4 forms is itself open [10]. Thus Hypothesis Z contains a known open problem as a special case. The heuristic density of non-ordinary primes is $\sim C_f/p$ (the Sato–Tate distribution on the normalized coefficients $a_p/p^{3/2}$ admits $\sim 4p^{1/2}$ multiples of p in $[-2p^{3/2}, 2p^{3/2}]$, each contributing mass $\sim p^{-3/2}$). The expected count up to x is thus $C_f \log\log x$ with $C_f \approx 1$. Finding 2 non-ordinary primes ($p = 11, 3137$) below 10^6 is consistent with $\log\log(10^6) \approx 2.9$. This is not a standard Lang–Trotter problem: the target set $\{a : p \mid a\}$ varies with p, and no analogue of Elkies' theorem (infinitely many supersingular primes) is known for weight ≥ 4.

Remark 32 (Evidence against Hypothesis Z)*.* The $2 \cdot \text{Poisson}(1/2)$ distribution (Remark 7 below) has unbounded support: extreme-value theory predicts $\max_{p \le N} Z(p) \sim 2\log\log N$. The current record $Z(p) = 12$ at $p = 159{,}977$ (with record progression $2, 4, 6, 8, 10, 12$) is consistent with this prediction. We therefore expect Hypothesis Z to fail: the first prime with $Z(p) = 14$ should appear near $p \approx 2 \times 10^7$ (the Gumbel prediction for the Poisson(1/2) pair maximum). A computation to $p = 5 \times 10^7$ would either refute Hypothesis Z or provide strong evidence for a structural cutoff.

Hypothesis 33 (Hypothesis $\bar{Z}$)**.** $\displaystyle\sum_{p \le x} Z(p) \ll \pi(x)$*, equivalently, the average of $Z(p)$ over primes $p \le x$ is bounded.*

Remark 34. Hypothesis $\bar{Z}$ is strictly weaker than Hypothesis Z and consistent with $Z(p) \sim 2 \cdot$ Poisson(1/2) (which gives $\mathbb{E}[Z(p)] = 1$, independent of p). It suffices for the conditional bound $\log G_n = O(\sqrt{n}(\log n)^{1+\varepsilon})$ (Theorem 41).

6 The denominator connection lemma

Lemma 35. *Let $p \ge 7$ be prime with $n/2 < p \le n$, and set $r = n - p$. Then $v_p(G_n) = 0$ if and only if $p \nmid b_r$.*

Proof. Since $p > n/2$, the only multiple of p in $\{1, \dots, n\}$ is p itself, so $v_p(d_n) = 3$.

Step 1. $v_p(d_n b_n) = 3 + v_p(b_n) \ge 3$, so $v_p(G_n) = 0$ requires $v_p(d_n a_n) = 0$, i.e., $v_p(a_n) = -3$.

Step 2 (variation of parameters). From Lemma 1, the Casorati identity telescopes to

$$\frac{a_n}{b_n} = \sum_{m=1}^{n} \frac{6}{m^3\, b_m\, b_{m-1}}. \tag{12}$$

For $p > n/2$, the only $m \in \{1, \dots, n\}$ with $p \mid m$ is $m = p$.

Step 3. By Theorem 4, $b_p \equiv b_0\, b_1 = 5 \pmod p$ (since p has base-p digits $(0,1)$), and by the symmetry $b_j \equiv b_{p-1-j}$, $b_{p-1} \equiv b_0 = 1 \pmod p$. Since $p \geq 7$, both are nonzero mod p.

The $m = p$ term in (12) has $v_p = -3$; all other terms have $v_p \geq 0$. Therefore

$$p^3 \cdot \frac{a_n}{b_n} \equiv \frac{6}{b_p\, b_{p-1}} \equiv \frac{6}{5} \pmod p.$$

Using $b_n \equiv 5\, b_r \pmod p$ (Theorem 4, since $n = 1 \cdot p + r$):

$$p^3\, a_n \equiv \tfrac{6}{5} \cdot 5\, b_r = 6\, b_r \pmod p.$$

Thus $v_p(a_n) = -3$ iff $p \nmid b_r$. □

Verification. For all $n \leq 300$ and primes $p \in (n/2, n]$ with $p \geq 7$: $v_p(G_n) = 0 \iff p \nmid b_{n-p}$ confirmed in 2101/2101 cases (100%).

Corollary 36 (Lucas reduction)**.** *For $p \geq 7$ and $p \in (n/2, n]$: $v_p(G_n) \geq 1$ iff $p \mid b_{n-p}$. By the Lucas congruence $b_n \equiv b_1\, b_{n-p} = 5\, b_{n-p} \pmod p$, this is equivalent (for $p \neq 5$) to $p \mid b_n$. Thus $B(n) := \#\{p \in (n/2, n], p \text{ prime} : v_p(G_n) \geq 1\}$ equals $\omega_{(n/2,n]}(b_n) + O(1)$, the number of* distinct prime factors of b_n lying between $n/2$ and n*, up to an absolute constant.*

The full conjecture $G_n = e^{o(n)}$ for all n is therefore equivalent to the *top-half radical estimate*:

$$\log \operatorname{rad}_{(n/2,n]}(b_n) = o(n). \tag{13}$$

Since $\log b_n \sim n \log(17 + 12\sqrt{2}) \approx 3.52\, n$, the height bound gives $\omega_{(n/2,n]}(b_n) \leq \log b_n / \log(n/2) = O(n/\log n)$, precisely one $o(1)$ factor short of the target. Computation for $n \leq 200{,}000$ gives $B(n) \leq 3$ and $B(n) = 0$ for 93.7% of values (see Remark 37 for the Poisson analysis).

Remark 37 (Random-model prediction)*.* Under a random-integer model, $\Pr(p \mid b_n) \approx 1/p$ for each prime p, so by Mertens' theorem

$$\mathbb{E}[B(n)] = \sum_{n/2 < p \leq n} \frac{1}{p} = \log\Big(\frac{\log n}{\log(n/2)}\Big) + O(1/\log n) \approx \frac{\log 2}{\log n} \ \to\ 0.$$

Computation for $n \leq 200{,}000$ confirms: $\mathbb{E}[B] \approx 0.065$, $\max B(n) = 3$, and the histogram matches $\text{Poisson}(\log 2/\log n)$ to high accuracy: $B(n) = 0$ for 93.7%, $B(n) = 1$ for 6.0%, $B(n) = 2$ for 0.20%, $B(n) = 3$ for 0.006% (the Poisson predictions with $\lambda = 0.065$ are 93.7%, 6.1%, 0.20%, 0.004%). By CRT-based approximate independence across primes, $B(n)$ should follow $\text{Poisson}(\log 2/\log n)$ with $\lambda \to 0$. Extreme-value theory then predicts $\max_{n \leq N} B(n) = O(\log N/\log\log N)$—trivially $o(n/\log n)$—so the random model *strongly* predicts the full conjecture. The obstruction is proving that b_n behaves "randomly" with respect to its prime factors in $(n/2, n]$, despite its highly structured factorization (supercongruences, Lucas decomposition, palindromic symmetry).

7 The unconditional density-one theorem

Theorem 38. *For a set of positive integers n of natural density 1, $G_n = e^{o(n)}$.*

Proof. Fix N large and consider $n \in (N, 2N]$. Write $B(n) = \#\{\text{bad primes } p > \sqrt{n} \text{ with } v_p(G_n) \geq 1\}$. By Proposition 3, the small-prime contribution is $O(\sqrt{n})$ for every n. Each bad prime contributes $v_p(G_n) \log p \leq 3 \log n$, so $\log G_n \leq O(\sqrt{n}) + 3B(n) \log n$. It suffices to show $B(n) = o(n/\log n)$ for all but $o(N)$ values of $n \in (N, 2N]$.

For $p > \sqrt{n}$, n has two base-p digits: $n = qp + r$ with $1 \le q \le \sqrt{2N}$ and $0 \le r < p$. By Theorem 4, $p \mid b_n$ iff $p \mid b_q$ or $r \in \mathcal{Z}_p$, where $\mathcal{Z}_p = \{j < p : p \mid b_j\}$. A prime p can be bad for n only if one of these holds.

Lower-digit incidence. For fixed $p \in (\sqrt{N}, 2N]$, an interval of N consecutive integers contains at most $N \cdot Z(p)/p + Z(p)$ values of n whose residue $r = n \bmod p$ lies in $\mathcal{Z}_p$. Set $\rho_p = Z(p)/p$. The total lower-digit incidence is

$$R_N \le \sum_{\sqrt{N} < p \le 2N} \big(N\rho_p + Z(p)\big).$$

By Proposition 9, $\rho_p \to 0$. Since $\pi(2N) = O(N/\log N)$:

$$\sum_p \rho_p = o(\pi(2N)) = o(N/\log N), \quad \sum_p Z(p) = \sum_p p\rho_p \le 2N \sum_p \rho_p = o(N^2/\log N).$$

Hence $R_N = o(N^2/\log N)$.

Leading-digit incidence. For fixed q, $p \mid b_q$ requires p to be a prime divisor of b_q, and $b_q \le (q+1) \cdot 64^q$ gives at most $O(q)$ such primes. For each pair (q, p), at most p integers in $(N, 2N]$ have $\lfloor n/p \rfloor = q$. Hence the total leading-digit incidence is

$$L_N \le \sum_{q=1}^{\lceil \sqrt{2N} \rceil} O(q) \cdot \frac{2N}{q+1} = O(N\sqrt{N}) = o(N^2/\log N).$$

Conclusion. Write $\sum_{n \in (N,2N]} B(n) \le R_N + L_N = o(N^2/\log N)$. For any fixed $\delta > 0$, Markov's inequality gives

$$\#\big\{n \in (N, 2N] : B(n) > \delta n/\log n\big\} \le \frac{o(N^2/\log N)}{\delta N/\log N} = o(N).$$

A dyadic decomposition shows the exceptional set has density zero, so $B(n) = o(n/\log n)$ for density 1. For all such n, $\log G_n = O(\sqrt{n}) + 3B(n)\log n = O(\sqrt{n}) + o(n) = o(n)$. □

Corollary 39 (Quantitative exceptional set)**.** *For every $\varepsilon > 0$,*

$$\#\{n \le N : \log G_n > \varepsilon\, n\} = O_\varepsilon(N^{2/3}).$$

Proof. We make the density-one argument quantitative using the explicit bound $Z(p) = O(p^{2/3})$ from Proposition 9. For $n \in (N, 2N]$, $\log G_n \le O(\sqrt{n}) + 3B(n)\log n$, so $\log G_n > \varepsilon n$ requires $B(n) > \varepsilon N/(7\log N)$ for N large.

Partial summation with $Z(p) \le Cp^{2/3}$ gives

$$\sum_{\sqrt{N} < p \le 2N} \frac{Z(p)}{p} \le C \sum_{p \le 2N} p^{-1/3} = \tfrac{3}{2}\, C\, \frac{(2N)^{2/3}}{\log(2N)} + O(N^{1/3}) = O\!\left(\frac{N^{2/3}}{\log N}\right).$$

The lower-digit incidence satisfies $R_N = N\sum Z(p)/p + \sum Z(p) = O(N^{5/3}/\log N)$ (the second sum, $\sum Z(p) \le C\sum p^{2/3}$, contributes the same order). The leading-digit incidence $L_N = O(N^{3/2})$ is dominated by R_N. Hence $\sum_{n \in (N,2N]} B(n) = O(N^{5/3}/\log N)$.

By Markov's inequality,

$$\#\big\{n \in (N, 2N] : B(n) > \varepsilon N/(7\log N)\big\} \le \frac{O(N^{5/3}/\log N)}{\varepsilon N/(7\log N)} = O(N^{2/3}/\varepsilon).$$

A dyadic decomposition over intervals $(N/2^{j+1}, N/2^j]$ gives the global bound, since $\sum_{j \ge 0}(N/2^j)^{2/3} = O(N^{2/3})$. □

Remark 40. The exponent 2/3 in the exceptional set is inherited directly from the exponent 2/3 in Proposition 9. More generally, if $Z(p) \le C\,p^{\alpha}$ for all primes, then the same argument gives $\#\{n \le N : \log G_n > \varepsilon n\} = O(N^{\alpha}/\varepsilon)$. In particular, the mesoscopic target $Z(p) = O(p^{1/2})$ (Remark 26) would yield an $O(N^{1/2})$ exceptional set, and Hypothesis Z ($Z(p) = O(1)$) would give $O(\log N \, \log\log N)$.

8 Counting bad primes: the conditional result

Theorem 41. *Assume Hypothesis $\bar{Z}$. Then for a set of positive integers n of natural density 1 and any function $\omega(n) \to \infty$,*

$$\log G_n = O\big(\omega(n)\sqrt{n}\log n\big).$$

In particular, $\log G_n = O(\sqrt{n}(\log n)^{1+\varepsilon})$ *for any $\varepsilon > 0$.*

Proof. As in Theorem 38, decompose $\log G_n = \sum_p v_p(G_n)\log p$. The small-prime contribution is $O(\sqrt{n})$. Split $B(n) = B_{\text{low}}(n) + B_{\text{lead}}(n)$, where B_{low} counts bad primes $p > \sqrt{n}$ with $n \bmod p \in \mathcal{Z}_p$ and B_{lead} counts those with $\lfloor n/p \rfloor \in \mathcal{Z}_p$.

Under Hypothesis $\bar{Z}$, set $M(t) = \sum_{p \le t} Z(p) \ll t/\log t$. The lower-digit incidence satisfies $R_N = \sum_{n\in(N,2N]} B_{\text{low}}(n) \le N \sum_{\sqrt{N}<p\le 2N} Z(p)/p + O(M(2N))$. Abel summation gives $\sum_{\sqrt{N}<p\le 2N} Z(p)/p = [M(t)/t]_{\sqrt{N}}^{2N} + \int_{\sqrt{N}}^{2N} M(t)/t^2\,dt \ll 1/\log N + \int_{\sqrt{N}}^{2N} dt/(t\log t) = O(1)$, so $R_N = O(N)$. By Markov, $B_{\text{low}}(n) \le \omega(n)$ for all but $O(N/\omega(N)) = o(N)$ values.

The leading-digit incidence satisfies $\sum_{n\in(N,2N]} B_{\text{lead}}(n) = L_N = O(N\sqrt{N})$ (as in Theorem 38). By Markov with threshold $\omega(N)\sqrt{N}$: $\#\{n \in (N,2N] : B_{\text{lead}}(n) > \omega(N)\sqrt{N}\} \le O(N/\omega(N)) = o(N)$.

For all remaining n: $\log G_n = O(\sqrt{n}) + 3[\omega(n) + \omega(n)\sqrt{n}]\log n = O(\omega(n)\sqrt{n}\log n)$. □

Remark 42 (The leading-digit bottleneck). The gap between the density-one bound $O(\omega(n)\sqrt{n}\log n)$ and the conjectured $O(\sqrt{n})$ comes from the leading-digit incidence $L_N = O(N\sqrt{N})$, which dominates the lower-digit incidence $R_N = O(N)$ under Hypothesis $\bar{Z}$. Heuristically, the lower digit $n \bmod p$ is uniform mod p with $B_{\text{low}}(n) = O(1)$, and the leading digit $\lfloor n/p \rfloor$ lands in $\mathcal{Z}_p$ with probability $\le C/p$, giving expected $B_{\text{lead}}(n) = \sum_p C/p = O(1)$. Making the leading-digit part rigorous—showing $B_{\text{lead}}(n) = O(1)$ pointwise or even for density 1—requires controlling the large-prime divisors of the Apéry numbers in short multiplicative intervals.

An exact divisor switch rewrites the leading-digit incidence as

$$B_{\text{lead}}(n) = \sum_{1\le q<\sqrt{n}} \#\{p \mid b_q : \max(\sqrt{n},\, n/(q{+}1)) < p \le n/q\}.$$

This reformulates the problem as a large-prime-divisor incidence problem for the sparse sequence (b_q).

For dyadic P, Q with $Q < P$, define

$$D(P,Q) = \#\{(p,q) : P < p \le 2P,\ Q < q \le 2Q,\ p \mid b_q\}.$$

Under Hypothesis $\bar{Z}$, $D(P,Q) \le \min(P, Q^2)/\log P$: the first term bounds the number of primes, the second counts prime divisors $> P$ of the product $\prod_{Q<q\le 2Q} b_q$. A dyadic decomposition of the hyperbola $pq \approx X$ gives $(1/X)\sum_{X<n\le 2X} B_{\text{lead}}(n) \ll X^{1/3}/\log X$, far from the conjectured $O(1)$ but nontrivial. The missing improvement is $D(P,Q) \ll Q/\log P$ (the random-divisibility prediction), which would yield a bounded dyadic mean.

Corollary 43. *The condition $Z(p) = o(p)$ is* necessary *for the conjecture: if $Z(p) \geq \varepsilon p$ for a positive proportion of primes, double-counting gives $\log G_n \geq c\varepsilon n$ for a positive proportion of n. Conversely, $Z(p) = o(p)$ implies $G_n = e^{o(n)}$ for density 1 (Theorem 38).*

Proof. For each prime $p \in (n/2, n]$ with $Z(p) \geq \varepsilon p$, Lemma 35 gives $v_p(G_n) \geq 1$ whenever $n-p \in \mathcal{Z}_p$. Suppose a proportion $\delta > 0$ of primes satisfy $Z(p) \geq \varepsilon p$. Summing over $n \in (N, 2N]$ and bad primes $p \in (n/2, n]$, the total incidence count is $\geq \delta\, \pi(2N) \cdot \varepsilon N/2 = \Omega(N^2/\log N)$. Dividing by N and applying Markov's inequality: a proportion $\geq \varepsilon\delta/4$ of integers n have $B(n) \geq \varepsilon\delta N/(4\log N)$, giving $\log G_n \geq 3B(n)\log(N/2) \geq c\varepsilon\delta\, n$. ☐

Remark 44 (The pointwise gap). The full conjecture $G_n = e^{o(n)}$ for *every* n is strictly stronger than the density-one statement. By Corollary 36, it is equivalent to the top-half radical estimate (13). The height bound gives $B(n) = O(n/\log n)$, but for an *adversarial* family (Z_p) with $|Z_p| \leq 1$, one may choose $Z_p = \{N-p\}$ for primes in $(N/2, N]$, producing $B(N) \geq \pi(N)-\pi(N/2) \sim N/(2\log N)$. Thus the needed pointwise bound $B(n) = o(n/\log n)$ requires arithmetic information specific to the Apéry sequence, not just the cardinalities $Z(p)$.

By Corollary 36, every bad prime $p > n/2$ satisfies $p \mid b_n$. Hence the conjecture implies $\log \operatorname{rad}_{(n/2,n]}(b_n) = o(n)$: the large prime radical of b_n must be subexponential, even though $|b_n| = e^{\Theta(n)}$.

9 Computational verification

We computed G_n exactly for $n = 1, \ldots, 1000$ using rational arithmetic.

n range	avg $\log(G_n)/n$	max $\log(G_n)/n$
1–50	0.276	0.93
51–100	0.108	0.26
101–200	0.086	0.18
201–300	0.048	0.10
301–400	0.038	0.08
401–500	0.024	0.05
501–600	0.029	0.06
601–700	0.017	0.04
701–800	0.014	0.04
801–900	0.016	0.04
901–1000	0.013	0.03

A power-law fit gives $\log(G_n)/n \sim 5\, n^{-0.87}$, i.e., $\log G_n \sim 5\, n^{0.13}$, far stronger than the density-one bound $o(n)$ of Theorem 38.

The exponent 0.13 should be interpreted with caution: if $\log G_n \sim \kappa \log n$, the effective log-log slope is $1/\log n = 0.145$ at $n = 1000$, close to the fitted 0.13. A random-gcd model—assigning each prime $p \leq n$ an independent probability $\sim 1/p$ of dividing G_n—predicts $\mathbb{E}[\log G_n] \sim \log n$ by the weighted Mertens theorem, with standard deviation $\sim (\log n)/\sqrt{2}$ of the same order. The data thus cannot yet distinguish $\kappa \log n$ from $n^{0.13}$; the logarithmic prediction is arithmetically more natural.

10 Remarks

1. The Beukers supercongruence $b_{mp^r} \equiv b_{mp^{r-1}} \pmod{p^{3r}}$ [4] provides tower rigidity: for each fixed prime, $v_p(b_n)$ is controlled by the base-p digits.

2. For the gap-2 polynomial $C_2(m) = P(m+1)$, the factorization $P(x) = (2x+1)(17x^2+17x+5)$ shows that gap-2 pairs can only occur at $m = (p-3)/2$ (when $2(m+1)+1 = 2m+3 \equiv 0$) or at roots of the quadratic $17x^2+17x+5$ (discriminant -51; exists over $\mathbb{F}_p$ iff $\left(\frac{-51}{p}\right) = 1$). Empirically, all observed gap-2 pairs arise from the linear factor at $m = (p-3)/2$.

3. For $1 \le j \le p-2$, the Apéry number $b_j \bmod p$ admits a character-sum representation as a Greene [8] Gaussian hypergeometric ${}_4F_3$ over $\mathbb{F}_p$ with parameter $\chi = \omega^j$ (Teichmüller character). However, the Weil bound controls the archimedean size $|b_j|_\infty = O(p^{3/2})$ but does *not* bound $Z(p)$, since zeros of b_j modulo p are a p-adic event. The gap polynomial argument above is specifically adapted to this p-adic zero-count problem.

4. Malik–Straub [11] studied Lucas congruences and zero-divisibility for all sporadic Apéry-like sequences; Rowland–Yassawi [14] proved automaticity of the Lucas property for diagonals of rational functions, which includes the Apéry sequence. Our $Z(p)$ data for the $\zeta(3)$ sequence agrees with their empirical findings.

5. In the standard toric coordinate, the truncated polynomial $H_p(t) = \sum_{j=0}^{p-1} b_j\, t^j \in \mathbb{F}_p[t]$ is the scalar Hasse–Witt invariant of the Apéry K3 pencil. Its *roots* in $\mathbb{F}_p$ are the nonordinary fibers; because the family has generic Picard rank 19, nonordinary implies supersingular at good primes. Our statistic $Z(p)$ counts vanishing *coefficients* of H_p, equivalently the vanishing finite Mellin coefficients $b_j = -\sum_{t\in\mathbb{F}_p^\times} H_p(t)\, t^{-j}$ for $1 \le j \le p-2$. Thus $Z(p)$ is a spectral sparsity statistic for the global Hasse/point-count function, *not* a count of supersingular fibers. Standard Hasse-stratification, Newton-polygon, and overconvergent F-crystal theory do not by themselves bound $Z(p)$; they control the root locus, not the Fourier support. A bounded-conductor realization of H_p as an ℓ-adic trace function could connect to the Forey–Fresán–Kowalski framework [7], but equidistribution of complex traces does not directly control exact zeros modulo p.

6. **Palindromic symmetry.** For $0 \le j \le p-1$ and $0 \le k \le p-1-j$, the identity $\binom{p-1-j}{k} \equiv (-1)^k\binom{j+k}{k} \pmod p$ holds (each of the k falling factors $p-1-j-i \equiv -(j+1+i)$). When $k \le j$, $\binom{p-1-j+k}{k} \equiv (-1)^k\binom{j}{k}$; when $k > j$, Lucas's theorem gives $\binom{p-1-j+k}{k} \equiv 0$ (since $p-1-j+k \ge p$ and $\binom{k-j-1}{k} = 0$). Therefore each term of $b_{p-1-j} = \sum_k \binom{p-1-j}{k}^2\binom{p-1-j+k}{k}^2$ with $k > j$ vanishes, and for $k \le j$ the $(-1)^{2k}$ signs cancel, giving $b_{p-1-j} \equiv b_j \pmod p$. The extra terms of b_j when $j > (p-1)/2$ (i.e., $k > p-1-j$) likewise vanish: $\binom{j+k}{k} \equiv \binom{j+k-p}{k} \equiv 0$ since $j+k-p < k$. Equivalently, $H_p(t)$ is palindromic: $t^{p-1} H_p(1/t) \equiv H_p(t)$. This explains the observed pairing: whenever $b_j \equiv 0$, also $b_{p-1-j} \equiv 0$, so zeros of b_j come in pairs $\{j,\, p-1-j\}$ (except at the self-paired center $j = (p-1)/2$, which vanishes only at non-ordinary primes). The "pair count" $K(p) = Z(p)/2$ (or $(Z(p)-1)/2$ at non-ordinary primes) follows Poisson(1/2) to the precision of the data (Proposition 27).

7. **Sym^2 squareness (theorem).** The Apéry differential operator is the symmetric square of the Picard–Fuchs operator of an underlying elliptic family (Stienstra–Beukers [16]). Caruso, Fürnsinn, Vargas-Montoya, and Zudilin [6] proved the mod-p squareness rigorously: for every prime $p \ge 5$, setting $\varepsilon_p = \frac{1}{2}(1 - \left(\frac{-6}{p}\right))$, there exists a unique (up to sign) polynomial $B_p \in \mathbb{F}_p[t]$ such that
$$H_p(t) \;=\; \Delta(t)^{\varepsilon_p} \cdot B_p(t)^2,$$
where $\Delta(t) = t^2 - 34t + 1$ is the conifold discriminant and B_p is squarefree and coprime to Δ. The proof uses the explicit identity $F(t(x)) = (1+x)\, h(x)^2$ between the Apéry and Franel

generating functions, the p-Lucas property of both sequences, and a quadratic-descent argument through the covering $\mathbb{F}_p(x)/\mathbb{F}_p(t)$ where $t = x(1-8x)/(1+x)$. The deck-transformation character $(-6/p)$ determines whether $\varepsilon_p = 0$ or 1:

$$\varepsilon_p = 0 \iff \left(\frac{-6}{p}\right) = 1, \qquad \varepsilon_p = 1 \iff \left(\frac{-6}{p}\right) = -1.$$

Equivalently, $\varepsilon_p = 0$ iff $p \equiv 1, 5, 7, 11 \pmod{24}$. These two cases occur with equal density 1/2 by Dirichlet's theorem.

Splitting the mean $Z(p)$ by residue class modulo 8 (and modulo 24) yields no significant difference (mean ≈ 1.00 in each class, $n \geq 2000$ primes per class up to 10^5); the squareness structure does not directly influence $Z(p)$, consistent with $Z(p)$ counting coefficient zeros rather than root zeros.

The factorization imposes a convolution structure: b_j is (up to the Δ^{ε_p} correction) the j-th coefficient of B_p^2, i.e., $b_j \equiv \sum_k \beta_k \beta_{j-k} \pmod{p}$ where β_k denotes the coefficients of B_p. The vanishing $b_j \equiv 0$ thus requires cancellation across $\sim p/2$ convolution terms. A Monte Carlo simulation (1000 random monic polynomials of degree $(p-1)/2$ over $\mathbb{F}_p$, for $p \in \{101, 503, 1009, 5003\}$) shows that the zero-count of the *squared* polynomial follows Poisson(1) (mean ≈ 1, variance ≈ 1, total variation distance to Poisson(1) equal to 0.018). However, the observed $Z(p)$ distribution over 667 primes $p \leq 5000$ is *not* Poisson(1) (TV = 0.43); instead it matches $2 \cdot$ Poisson(1/2) with TV = 0.023. The explanation is the palindromic pairing: since $b_{p-1-j} \equiv b_j$, zeros come in pairs $\{j, p-1-j\}$ and $Z(p)$ is almost always even. Denoting by $Z_{\text{pair}}(p)$ the number of zero pairs, the data show $Z_{\text{pair}} \sim$ Poisson(1/2), so $Z(p) = 2Z_{\text{pair}}(p) \sim 2 \cdot$ Poisson(1/2). Thus squareness explains the $O(1)$ mean, while palindromic symmetry supplies the parity constraint. The Poisson tail predicts $\max_{p \leq N} Z(p) \sim 2 \log\log N$ via extreme-value theory, strongly suggesting that $Z(p)$ is unbounded (i.e., Hypothesis Z is false). Whether the algebraic provenance of B_p imposes a bounded *average* (Hypothesis $\bar{Z}$) remains the central open question.

8. **Toric framework.** The Laurent polynomial $f = w(x + x^{-1} + y + y^{-1} + z + z^{-1})$ on $(\mathbb{F}_p^\times)^4$ has Newton polytope $\Delta = \text{conv}(\{0\} \cup \text{Supp}(f))$, a 4-dimensional pyramid over the regular octahedron in $\mathbb{R}^3$. Its f-vector is $(7, 18, 20, 9)$, its normalized volume is $4! \cdot \text{vol}(\Delta) = 8$, and it is Kouchnirenko nondegenerate for every prime $p \geq 5$: for each of the 27 nonempty faces σ not containing the origin, the toric critical system $\{x_i \partial f_\sigma / \partial x_i = 0\}$ has no solution in $(\mathbb{F}_p^\times)^4$. For the Adolphson–Sperber twisted Hodge polygon with w-Kummer twist $\chi_w = \omega^{-j}$ (character parameter $a = j/(p-1)$), the slopes are a, $a+1$, $a+2$, $a+3$ with multiplicities $1, 3, 3, 1$, confirming that the first slope equals a for every $a \in \{k/100 : 0 \leq k \leq 99\}$. By Newton-above-Hodge (Adolphson–Sperber [2]), the p-adic Newton polygon lies on or above this Hodge polygon, and ordinarity (Newton = Hodge) implies the existence of a unit root. The $Z(p)$ bound thus reduces to proving ordinarity for all but $O(1)$ twist parameters j.

9. **Branch obstruction.** The $p-1$ Kummer twists χ^{-j} ($j = 0, \ldots, p-2$) are distinct characters of conductor p. In Iwasawa-theoretic coordinates $\chi^j = \omega^a \langle\cdot\rangle^s$ with $a = j \bmod (p-1)$, the horizontal family $\{j : 0 \leq j < p-1\}$ hits *each* of the $p-1$ branches ω^a exactly once. Weierstrass preparation and bounded-λ rigidity control zeros *within one branch* (the vertical direction j, $j + (p-1)$, $j + (p-1)p, \ldots$, where Dwork congruences live) but not *across* branches. Thus there is no p-adic analytic function of one variable whose zero count majorizes $Z(p)$. Any "rigidity" proof of Hypothesis $\bar{Z}$ would need a mechanism not supplied by existing Iwasawa–Dwork theory.

10. Any estimate $\log G_n = o(n)$—whether $O(\log n)$, $O(\sqrt{n})$, or $O(n^{0.13})$—yields $\gamma = 0$ in the exponent $G_n = e^{\gamma n + o(n)}$ and therefore does not improve the Rhin–Viola irrationality measure $\mu(\zeta(3)) \leq 5.513$ [15], which depends only on the exponential growth rate $|b_n\zeta(3) - a_n| = e^{-(\log\alpha + o(1))n}$. An improvement would require an exponentially large common divisor, which is not predicted by any current model.

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